

What is an element?

An element is a substance that cannot be refined or purified into simpler ingredients. Elements are made of building blocks called atoms and each element has its own unique set of atoms. Everything in the Universe is made from elements, either in their pure form, or combined together to make new substances called compounds.

Classical elements

The idea of an element is very old. In ancient cultures, people believed that all things were made from mixtures of just four elements: earth, water, air, and fire. They thought that hot and dry things contained fire and air, while cold and wet things were made of earth and water.



Earth



Water



Air



Fire

Modern elements

The scientific study of the properties and reactions of elements is called chemistry. This has found that there are at least 118 elements. Most elements are created inside a star or in a supernova – the explosive ending of a big star's life. Scientists have also been able to create the heaviest elements in laboratories.

The Crab Nebula, the remains of a supernova, is rich in hydrogen.

